ADMINISTRATOR & OPERATOR USER GUIDE

SmartAttendance™ Touchscreen Terminal

Complete Step-by-Step Instructions for Classroom Attendance, Student Registration, Lecture Scheduling via Telegram, Battery Charging, Reports & Over-The-Air Updates

- **Simple Touchscreen Operation:** 7-inch color display designed for intuitive everyday classroom use.
- **Fast & Hygienic Attendance:** Sub-second contactless facial verification with on-screen visual confirmation.
- **Works Anywhere:** Operates 100% offline without internet; optional Wi-Fi syncs records seamlessly.
- **Dedicated Registration:** Enroll students via the on-device Enroll app or from your smartphone via Wi-Fi.
- **Automated Class Scheduling:** Lecturers schedule classes easily through the official Telegram Bot.
- **Fail-Safe Updates (OTA):** Dual-bank rollback architecture ensures software updates are safe and painless.
- **Portable Battery Power:** Built-in rechargeable battery provides 4 to 6 hours of continuous scanning.
- **Instant Reports:** Export attendance directly to an Excel-compatible CSV file on the memory card.

Target Audience:	Course Lecturers, Department Officers, School Admins	**Edition:**	Non-Technical / Layman Operator Guide
Device Model:	SmartAttendance 7.0″ Touch Terminal	**Telegram Bot:**	@MyUniAttendance_Bot
Typography:	Times New Roman (12pt Body)	**Publication Date:**	October 2026

[v] BEST PRACTICE / TIP: About This Manual

This guide is written specifically for day-to-day operators. It explains how to operate the slide switch, check battery colors, charge the device, enroll students, run attendance sessions, schedule classes, export reports, and perform safe OTA updates. For circuit schematics and hardware repair guides, refer to the separate Technician Manual.

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Quick Overview: How the System Works

The SmartAttendance terminal is an all-in-one smart device designed to make taking attendance fast, accurate, and completely paperless. Here is how simple the daily routine is:

- **1. Bring to Class:** Place the device on the lecturer's desk or mount it by the classroom entrance.
- **2. Power On:** Slide the power switch on the left side **UP**. The device boots in about 5 seconds.
- **3. Automatic Session:** If the class was scheduled via Telegram, the device automatically shows the active course name on screen. If not, tap the screen to select your course.
- **4. Students Walk By:** Students pause for just one second in front of the screen. A green box and green visual confirmation banner appear confirming their name.
- **5. Instant Records:** Attendance is saved immediately into the memory card. Students receive a Telegram alert on their phones, and the lecturer gets a complete attendance sheet after class.

Chapter 1: Getting to Know Your Terminal

Familiarize yourself with the exterior parts and verified physical controls of the terminal:

Component	Location	Operational Function & Instructions
Power Slide Switch	Left Side (Upper)	Mechanical slide switch: **Slide UP to power ON**. **Slide DOWN to power OFF**. Provides instant hardware power control.
USB-C Charging Slot	Left Side (Below Switch)	Standard USB Type-C port for charging and power. **Recommended power rating: 5V, 2A to 3A**. Compatible with standard phone fast chargers.
Camera Lens	Top of Screen	Integrated high-definition camera facing outward toward students. Keep the lens clean and free of fingerprints.
7.0″ Touchscreen	Front Face	Crisp 7-inch color display. Shows live camera view, visual confirmation banner, camera profiles, clock, and battery level.
Memory Card Slot	Bottom Slot	Houses the MicroSD card holding student records, attendance logs, and AI models. **CRITICAL: DO NOT REMOVE!** Removing the card halts attendance functions.
Desktop Launcher	Home Display	Grid of 7 dedicated apps: **Scanner** (attendance), **Enroll** (new user +), **Users** (manage profiles), **Reports**, **Settings**, **Files**, and **Schedules**.

Chapter 2: Powering On, Battery Levels & Charging

- **Turning On:** Slide the switch on the left side **UP**. The screen lights up immediately and loads the main scanner in ~5 seconds.
- **Turning Off:** Slide the switch on the left side **DOWN** to completely power off the unit.
- **Charging the Device:** Plug a certified **5V 2A–3A USB Type-C charger** into the charging slot on the left side (located right below the power switch). A full charge takes approximately **2.5 to 3.5 hours**.
- **Continuous Power:** You can keep the USB-C charger connected during long lecture days. The device supports pass-through charging without battery degradation.

Understanding the Battery Color Indicators (No Bars / No Green)

The battery status in the top-right corner uses clean **numerical percentages and color indicators** (no multi-colored bars, and no green indicator):

Status / Mode	Screen Display	Color Indicator	What It Means & What You Should Do
Charging Connected	Lightning Bolt Icon + Percentage (e.g. [bolt] 74%)	**Glowing Cyan Blue (#00CFFF)**	The device is actively charging from the USB-C slot. Both the lightning bolt icon and percentage text glow in bright cyan blue.
Normal Battery (15% – 100%)	Battery Icon + Percentage (e.g. 88%)	**White** (Dark Theme) / **Black** (Light Theme)	Standard discharging mode. Battery is healthy and ready for active attendance scanning. (Lasts ~4 to 6 hours continuous).
Low Battery (< 15%)	Battery Icon + Percentage (e.g. 12%)	**Vivid Red (#FF4444)**	Both the icon and percentage turn solid red. A notification banner warns that the battery is low. Plug in the USB-C charger soon.

Critical Cutoff (5%)	Flashing Red Alert	**Red / Auto-Protect**	**At 5%, the terminal automatically saves all attendance records to the memory card and safely shuts down** to prevent data loss.

Chapter 3: First-Time Setup Wizard

When you power on a fresh terminal, a simple 4-step wizard welcomes you:

- **Step 1: Set Your Master Admin PIN:** Choose a secure **4 to 8-digit PIN** (default initial code is 1234). This PIN protects student registration, roster management, reports, and settings from unauthorized tampering. Do not share this PIN with students.
- **Step 2: Enter School / Department Details:** Type your Department Name (e.g., *Computer Science*) and Terminal Location (e.g., *Lecture Hall B*). These names appear on all exported attendance reports.
- **Step 3: Set Date & Time:** Confirm the current time and date. If you connect to Wi-Fi, the device sets its clock automatically over the internet.
- **Step 4: Connect to Campus Wi-Fi (Optional):** Select your Wi-Fi network and type the password. If your classroom does not have Wi-Fi, tap **[Skip]** — the device works 100% offline!

Chapter 4: Daily Attendance Scanning in the Classroom

During class, the device runs in automatic scanning mode. Taking attendance is completely contactless and visual (the system does not use audio or sound chimes):

- **1. Student Positioning:** The student approaches the terminal and pauses about **2 to 3 feet** (arm's length) in front of the screen, looking toward the camera lens at the top of the display.
- **2. Sub-Second Facial Verification:** The device tracks the face with a green reticle box and verifies their identity in under 1 second.
- **3. Understanding Visual Screen Confirmations:**
  - **GREEN RETICLE & GREEN BANNER: Success!** The bottom feedback banner turns deep green with: "Welcome, [Student Name]! Recorded at 09:15 AM".
  - **ORANGE BANNER: Duplicate Scan:** Shows "Already Recorded: [Student Name]" if scanned again within 15 minutes.
  - **RED TEXT ('Unknown'):** The face is not registered in the system, or the student needs to step closer.
- **4. Helpful Guidance Prompts on Screen:**
  - *"Please close mouth & look straight"* — student is laughing, talking, or yawning.
  - *"Glare detected — adjust angle from light"* — sunlight or a bright lamp is reflecting into the lens.
  - *"Blurry image — please hold still"* — student is walking past too quickly.
- **5. Camera Profile Buttons (Right Panel):** If classroom lighting changes, tap any of the 5 quick profile buttons on the right side of the screen: **Auto**, **Outdoor**, **Indoor**, **Bright**, or **Low Light** for instant exposure optimization.

[v] BEST PRACTICE / TIP: Selecting an Unscheduled Lecture

If an extra or unscheduled class takes place, tap **[Select Course]** on the top banner, enter your 4 to 8-digit Admin PIN, and pick the course from the drop-down list. All scans will immediately be filed under that course.

Chapter 5: Enrolling New Students & Staff

Registration only needs to be done once per student. The system provides two flexible enrollment methods:

Understanding the Desktop Launcher Apps

The terminal home screen features 7 dedicated applications in an intuitive grid: **Scanner** (live attendance), **Enroll** (new student registration), **Users** (manage registered roster), **Reports** (attendance logs & CSV), **Settings**, **Files**, and **Schedules**.

Important Note: The **Users** app is strictly for viewing existing student profiles, re-enrolling faces when appearance changes, or deleting records. **There is no 'Add User' button inside the Users app.** To register a new student directly on the device, always tap the dedicated **Enroll** app (marked with a green + symbol) from the desktop launcher!

Method 1: Directly on the Terminal Screen (Enroll App)

- From the main desktop launcher, tap the **Enroll** app (green + icon).
- Enter your **4 to 8-digit Master Admin PIN** to authorize the registration session.
- Type the student's **Full Name**, **Matriculation / Student ID** (e.g., CSC/2022/1045), and optional **Phone Number** (e.g., 08012345678). Select their Role (Student or Lecturer).
- Tap **[Start Face Capture]**. Guide the student to stand in front of the camera (2–3 feet away) with a natural, relaxed face.
- The screen automatically captures 30 high-speed frames while the progress bar advances from 0% to 100%.
- A **Quality Scorecard** appears (e.g., 89/100 EXCELLENT). Tap **[Save]** to finalize registration.

Method 2: From Smartphone / Laptop (Wi-Fi Portal)

During large orientations or busy matriculation days, you can register students from your phone or laptop while the terminal scans them automatically:

- **1. Connect to Terminal Wi-Fi:** On your mobile phone, tablet, or laptop, open Wi-Fi settings and connect to: **Attendance_Setup** (open network, no password required).
- **2. Automatic 'Sign into network' Notification:** Most modern smartphones (iOS, Android) and laptops will automatically pop up a system prompt stating **'Sign into network'** or 'Log in to Wi-Fi'. Tap this prompt to launch the registration portal (register.html) directly—**there is no need to manually type any IP address!**
- **3. Manual Fallback:** If your device does not display the automatic prompt, open any web browser (Chrome, Safari, Edge) and enter: **http://192.168.4.1/register.html**.
- **4. Enter Dynamic 6-Digit Session PIN:** Enter the **dynamic 6-digit session PIN** (e.g., 834921) displayed prominently in the banner at the top of the terminal screen. This PIN is generated afresh for each enrollment session to prevent unauthorized student submissions.
- **5. Submit Student Details:** Type the student's Name, Matriculation ID, Role, and Phone Number, then tap **[Queue for Capture]**.
- **6. Contactless Capture:** The student steps in front of the terminal camera; the device scans them automatically and displays instant confirmation on both the terminal screen and your phone!

Understanding Enrollment Quality Scores

Quality Rating	Score Range	What It Means & What You Should Do
EXCELLENT	85 to 100	Optimal face scan. High accuracy under all indoor lighting conditions. Saved automatically.
GOOD	70 to 84	Very reliable face scan. Suitable for normal daily classroom verification.
AVERAGE	55 to 69	Minor blur or slight head tilt during capture. Accepted, but student should look straighter.
POOR	Below 55	Too blurry, dark, or angled. **Tap the 'Redo Enrollment' button** to re-scan under better posture.

Chapter 6: Managing Registered Students

Administrators can inspect, search, update, or remove registered users at any time using the **Users** app on the desktop launcher:

- **Opening the Users App:** From the desktop launcher, tap the **Users** icon (list symbol) and enter your **4 to 8-digit Master Admin PIN**.
- **Searching Students:** Use the top search bar to find any student instantly by typing their full name or matriculation number.
- **Updating a Face Scan (New Hair, Beard, or Glasses):** If a student significantly changes their appearance and recognition confidence drops, find their profile card and tap the orange **[Re-enroll Face]** button. This updates their facial scan **without losing their past attendance history or student ID!**
- **Removing a Student:** If a student drops a course or graduates, tap the red **[Delete User]** button and confirm. Their face scan is removed, but historical attendance records remain safely archived for university audits.

Chapter 7: Scheduling Lectures via Telegram

Instructors can schedule their classes anytime from their phone using the official Telegram Bot: **@MyUniAttendance_Bot**.

Step 1: One-Time Lecturer Verification

1. Open Telegram on your phone and search for @MyUniAttendance_Bot.
2. Tap **Start** (or send /start).
3. Tap the button **[Share Contact Details]**. The bot automatically links your phone number to your lecturer profile.

Step 2: Scheduling a Lecture in 5 Steps (`/schedule`)

Send /schedule to the bot. The bot will guide you through five quick questions:

Step	Bot Asks For	Example Answer	Helpful Guidance
1	Course Code	CSC 301	Type your department course code.
2	Course Title	Database Systems	Type the full name of the subject.
3	Lecture Date	24/10/2026	Format: Day/Month/Year (DD/MM/YYYY).
4	Start Time	09:00	Format: Hour:Minute in 24h time (e.g. 09:00, 14:00).
5	End Time	11:00	Format: Hour:Minute when lecture finishes.

What Happens Next? When the terminal connects to Wi-Fi, it automatically downloads your schedule. When class time arrives, the device automatically sets up the session and begins taking attendance under your course code!

Chapter 8: Viewing & Exporting Attendance Reports

You can inspect class attendance immediately on screen or save it to an Excel-friendly file:

- **Viewing On-Screen Reports:** Open the desktop launcher, select the **Reports** app, and enter your **4 to 8-digit Admin PIN**. Choose today's date or select a course code to see the total number of students registered, present count, absent count, and attendance percentage.
- **Exporting to Excel (CSV File):** Tap the blue **[Export CSV to SD]** button. The device writes an Excel-compatible spreadsheet file directly to the memory card under /sdcard/reports/.
- **Accessing Exported Files:** Connect the terminal via USB to your computer (or access via the web portal) and open the file in Microsoft Excel, Google Sheets, or WPS Office. The sheet shows Student Name, Matric Number, Course Code, Exact Verification Time, and Status (PRESENT).
- **Automatic Telegram Class Summary:** When the lecture session ends, the Telegram bot automatically sends the lecturer a message with total attendance numbers and attaches the full CSV roll sheet directly to their Telegram chat!

Chapter 9: System Settings & Personalization

Open the **Settings** app from the desktop launcher (enter your 4 to 8-digit Admin PIN) to customize terminal operation:

Setting	How to Adjust	Recommendation for Operators
Screen Brightness	Drag the Brightness slider (10% to 100%)	Keep at 60% – 70% indoors to save battery while keeping the display clear.
Theme Switch	Toggle the Light / Dark theme switch	Dark theme is default (optimal for indoor halls and reduced battery power draw).
Wi-Fi Setup	Tap [Scan Networks], pick SSID, enter password	Connect to campus Wi-Fi so reports, schedules, and OTA updates sync automatically.
Change Admin PIN	Tap [Security] -> [Change PIN]	Change the initial default PIN (1234) to a confidential **4 to 8-digit code** known only to authorized staff.

[i] HELPFUL NOTE: Memory Card Protection

The MicroSD card inserted into the bottom slot contains the entire offline student database and facial recognition models. Never remove the memory card during operation. If the card is removed, the device will display an error and halt scanning until re-inserted.

Chapter 10: Software & Firmware Updates (Over-The-Air / OTA)

The SmartAttendance terminal includes a state-of-the-art **Over-The-Air (OTA) update system**. OTA updates allow university administrators to install new software features, recognition model improvements, and security patches without needing to disassemble the device, attach programming cables, or risk breaking settings.

Enterprise Fail-Safe Protection (Dual-Bank A/B Rollback)

Unlike ordinary consumer gadgets that can get 'bricked' if power cuts out during an update, the SmartAttendance terminal uses an enterprise-grade **dual-bank flash memory system** (Partition Bank A: ota_0 and Bank B: ota_1):

• **Zero Risk of Bricking:** When an update is installed, it is written into the inactive standby bank while the current bank remains completely intact.

• **Automatic Diagnostic Self-Test:** When the device reboots, it tests the new firmware. If the new software encounters any crash or error, the internal bootloader **automatically cancels the update and immediately boots back into the previous working version!** Your terminal will never fail to start.

Method 1: Local Update via Web Portal (`/update`)

- **1. Connect to Terminal:** Connect your computer or smartphone to the terminal's Wi-Fi network (Attendance_Setup) or ensure both are on the same campus Wi-Fi network.
- **2. Open Update Page:** In your web browser, navigate to: **http://<device-ip>/update** (or http://192.168.4.1/update when on Attendance_Setup).
- **3. Authenticate:** Enter your **4 to 8-digit Master Admin PIN** to authorize firmware flashing.
- **4. Select Firmware File:** Tap **[Choose File]** and select the official SmartAttendance.bin update file supplied by your IT department.
- **5. Install & Restart:** Click **[Upload & Install]**. A progress bar tracks the update. Once flashing reaches 100%, the terminal verifies integrity and automatically restarts into the new version in ~10 seconds.

Method 2: Remote Update via Telegram Bot (`/ota`)

Authorized department administrators can push firmware updates remotely using Telegram:

1. Open @MyUniAttendance_Bot in Telegram from your verified administrator account.
2. Send the command: **/ota**.
3. When the terminal is connected to campus Wi-Fi, it downloads the verified update securely in the background and prompts the operator on screen to confirm reboot.

[v] BEST PRACTICE / TIP: Checking Your Firmware Version

To check your terminal's current software version, open the **Settings** app from the desktop launcher and tap **About Device**. The screen displays the current firmware version, build date, and active bank (Bank A or Bank B).

Chapter 11: Simple Troubleshooting & FAQs

Here are answers to the most common questions and simple fixes for everyday classroom situations:

Problem Encountered	Common Cause	Easy Solution for the Operator
Device will not turn on	1. Switch is DOWN. 2. Battery is drained.	Ensure the slide switch on the left side is pushed **UP**. If battery is dead, plug in the 5V 2-3A USB-C charger below the switch.
Student's face shows Red ('Unknown')	1. Not enrolled. 2. Changed appearance. 3. Heavy sunglasses/hat.	Ask student to remove sunglasses or low hats. If they changed hairstyle or grew a beard, use **Re-enroll Face** in the Users app.
Screen says 'Glare detected'	Direct sunlight or lamp shining into lens.	Turn the terminal slightly away from sunny windows, or tap the **Bright** / **Outdoor** camera profile button on the right panel.
Student says they didn't get counted	Scanned again within 15 minutes.	Explain that the 15-minute duplicate prevention stops double check-ins. Check Attendance Reports to confirm they are already marked PRESENT.
Wi-Fi icon has an 'X' or shows disconnected	Out of Wi-Fi range or router restarted.	**Do not worry!** The device works 100% offline. Attendance continues normally and syncs automatically when Wi-Fi reconnects.
Touchscreen is not responding smoothly	Fingerprints, moisture, or dust on glass.	Gently wipe the glass screen with a clean, dry microfiber cloth. Avoid wet paper towels or harsh window cleaners.

Everyday Care & Maintenance Tips

- **Screen Cleaning:** Always use a dry, soft microfiber cloth to clean the glass touchscreen and camera lens. Never spray water, alcohol, or liquid cleaners directly onto the unit.
- **Safe Storage:** When not in use, slide the power switch DOWN and store the terminal in a clean, dry office drawer or padded bag. Keep away from excessive dust or extreme heat.
- **Charging Safety:** Use a certified 5V 2A–3A USB Type-C power adapter. Do not leave the device charging on soft beds or enclosed cushions.
- **Memory Card Slot:** Keep the MicroSD memory card inserted in the bottom slot at all times.

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